



Visvesvaraya National Institute of Technology, Nagpur Electronics and Communication Engineering Department

Executive M.Tech. in Applied AI & Communication

Cohort- 3A & 3B, Sem-3

Eval. 1 (05/04/2026)

NATURAL LANGUAGE PROCESSING LABORATORY (ECL-545)

Automated Analysis of Meter Testing Remarks

Presented by-
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Problem Statement & Dataset

❑ Output Labels (Action Categories)

We mapped remarks into structured classes:

- TESTED_OK → No further action required
- INSTALLATION_REPLACEMENT → Meter replacement required
- CEG_ANALYSIS → Seal / technical investigation cases
- ZONE_RECTIFICATION → Network / voltage issues
- SITE_NOT_READY → Access / safety issues
- ADMIN_CLOSURE → Administrative closure cases
- CUSTOMER_DEPENDENT → Customer unavailable / refusal
- VISIT_REQUIRED → Revisit or rescheduling required

❑ Key Insight:

Dataset converts:

field-level textual remarks → structured operational decisions.

❑ Dataset Description:

- Source: Real-time utility meter testing records (1-year data)
- Total Dataset Size: ~**15,000 manually entered remarks**

❑ Original Dataset Structure

- 221 total columns available in raw dataset (150+ MB in size)

❑ Selected Relevant Columns (Final Use Case)

We filtered and used only:

- Serial Number → Reference ID
- Text Remarks → Input Feature (NLP Text)
- Hold Cancel Reason
- Hold Cancel Remarks
- Current Status → Target-related information

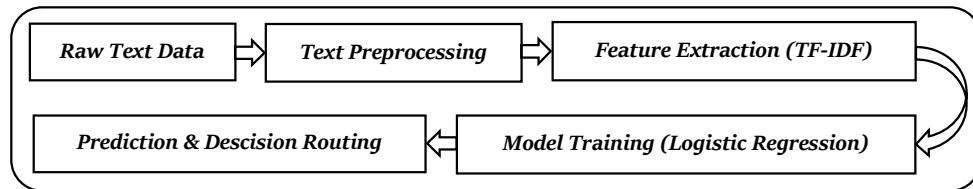
Methodology and Algorithms

❖ Problem Formulation:

The task is formulated as a Supervised Multi-Class Text Classification Problem

Map unstructured meter testing remarks → predefined department/action classes

❖ Project Pipeline:



➤ Text Preprocessing Techniques:

To transform raw text into a clean, standardized corpus for feature extraction. We applied standard NLP normalization to reduce noise and improve feature quality:

- **Case Normalization** → Lowercasing for uniformity
- **Noise Removal** → Elimination of punctuation, digits, and special characters
- **Stopword Removal** → Removal of high-frequency non-informative tokens
- **Tokenization** → Conversion of text into individual lexical units
- **Lemmatization** → Morphological normalization (e.g., running → run)

➤ Feature Engineering: TF-IDF Vectorization:

Used Term Frequency-Inverse Document Frequency for feature representation. Mathematically,

$$TF-IDF = TF(t,d) \times \log(N / DF(t))$$

where:

- $TF(t,d)$: Term frequency in document
- $DF(t)$: Document frequency
- N : Total number of documents

4/1/2026

➤ Model Selection: Logistic Regression:

Used **Logistic Regression for classification**. Estimates probability using Sigmoid Function:

$$P(y) = 1 / (1 + e^{(-z)})$$

Extended using Softmax Function for multi-class classification

Reason why Logistic Regression is Chooosed?

- Efficient for high-dimensional sparse data (TF-IDF vectors)
- Performs well in linear decision boundary problems
- Computationally efficient and scalable
- Provides probabilistic outputs for class prediction

➤ Model Training Strategy:

- Dataset split into:
 - ✓ Training Set (80%)
 - ✓ Testing Set (20%)
- Optimization using **Maximum Likelihood Estimation**
- Model Training will be Evaluated using:
 - ✓ Accuracy
 - ✓ Precision
 - ✓ Recall
 - ✓ F1-Score

➤ Prediction & Decision Routing:

Predicted class corresponds to:

- ✓ Concerned Department
- ✓ Further mapped to Actionable Workflow

➤ Live Deployment & User Testing:

- After achieving satisfactory performance during model training and testing, the system is deployed on a **new/live dataset of meter testing remarks**.
- The trained model processes real-time user inputs and automatically classifies remarks into the appropriate department, enabling **direct routing without manual intervention**.

Work Done till now

1. Dataset Preparation:

- Collected and structured meter testing remarks dataset
- Defined target classes corresponding to **concerned departments/actions**
- Ensured proper labeling for supervised learning

2. Text Preprocessing & Cleaning:

Performed multiple NLP-based cleaning steps to improve data quality:

- Converted text to **lowercase** for uniformity
- Removed **punctuation, numbers, and unwanted symbols**.
- Eliminate **noise patterns and irrelevant text**.
- Applied **stopword removal** to discard non-informative words.
- Performed **tokenization** to split text into meaningful units.
- Applied **lemmatization** to normalize words to root form.

3. Feature Engineering:

- Implemented **Term Frequency-Inverse Document Frequency (TF-IDF)** vectorization
- Transformed textual data into **numerical feature vectors**
- Captured importance of domain-specific keywords

STEP 1 – Cleaning + Lowercase (example)

```
site unsafe for meter testing discuss with  
consumer sh ravinder dabas
```

STEP 2 – Tokenization + Stopword Removal (example)

```
['site', 'unsafe', 'meter', 'testing',  
'discuss', 'consumer', 'sh', 'ravinder',  
'dabas']
```

STEP 3 – TF-IDF Vectorization (example)

```
site: 0.247  
unsafe: 0.321  
meter: 0.055  
testing: 0.068  
discuss: 0.334  
consumer: 0.268  
sh: 0.356  
ravinder: 0.484  
dabas: 0.533
```

Results Obtained till now

[Step-1] Initial Data Structuring & Classification Setup

NLP Output Label	Codes	Description	Next Concerned Department
TESTED_OK	—	No further action	None / Case Closure System
INSTALLATION_REPLACEMENT	D255, D490	Meter replacement required	Meter Replacement / Installation Team
CEG_ANALYSIS	D168, D320, D460	Seal missing / suspected / technical analysis	CEG / Enforcement / Vigilance Team
ZONE_RECTIFICATION	D395, D510, D330, D400	Voltage / network issue	Network / Zone Maintenance Team
SITE_NOT_READY	D030, D315, D430, D440, D500, D271, D606, D603, D275, D601	Locked / no access / unsafe	Field Operations / Meter Testing
ADMIN_CLOSURE	D120, D385, D610	Wrong notification / work already done	Back-Office / Administrative Operations
CUSTOMER_DEPENDENT	D560	Consumer not available / refused	Customer Service / Call Center
VISIT_REQUIRED	D270, D360, D570, D580	Visit not done / reschedule	Scheduling / Field Dispatch Team

[Step-2] Text Preprocessing & Cleaning

SR_NO	TEXT_REMARKS	ACTION_CATEGORY	HOLD_CANCEL_REASON
710	Meter already replaced New meter No -79039984	Meter already replaced New meter No	D270
1860	Meter display showing temper and MDI 347.98 kw suspected referred to enforcement and discuss with consumer sh Hament on phone no 9873511623	Meter display showing temper and MDI kw suspected referred to enforcement and discuss with consumer sh Hament on phone no	D320
2036	meter already removed at site date on 5/11/25 consumer name is Suresh mobile number is 9891816286	meter already removed at site date on consumer name is Suresh mobile number is	D360
2582	3 time visit but no person at site no response on call -9811304502	time visit but no person at site no response on call	D385
3375	2 time visit but premises found locked no response on call - 9899460445	time visit but premises found locked no response on call	D430
8421	EL LED found Glowing even on External Load (2 KW), hence meter is defective and therefore same needs to be replaced under defective category Contacted Person- Jagbir Singh (9891593144)	EL LED found Glowing even on External Load KW hence meter is defective and therefore same needs to be replaced under defective category Contacted Person Jagbir Singh	D490

Results Obtained till now

[Step-3] TF-IDF Feature Engineering

✓ Key Insights:

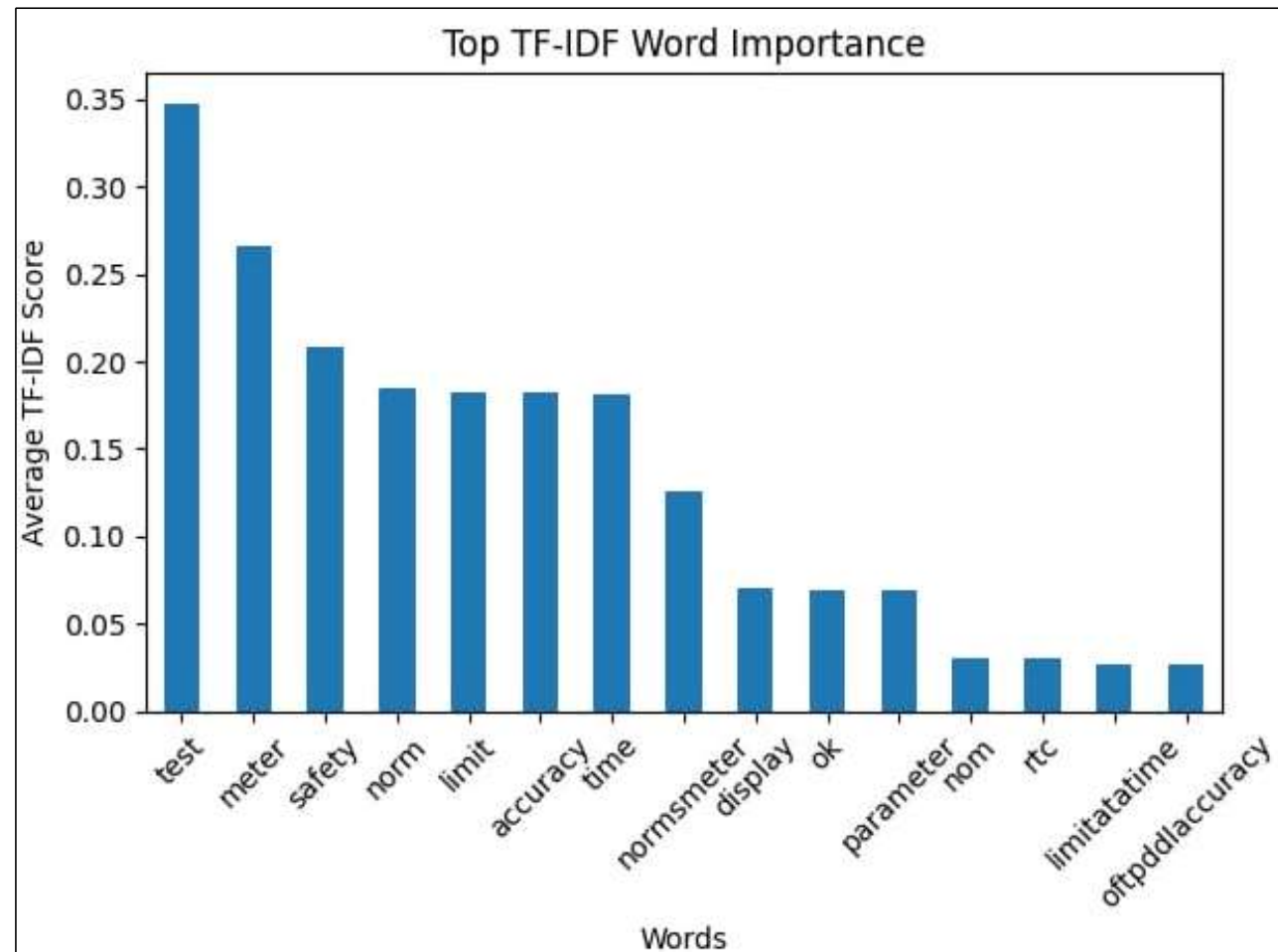
- Generated **1048 unique textual features** using TF-IDF vectorization
- Each word assigned a numerical importance score based on its relevance
- Successfully transformed unstructured text into meaningful numerical representation

✓ Feature Importance Analysis (Graph Insight):

- The graph shows **Top TF-IDF** weighted words influencing classification
- Words like: **“test”, “meter”, “safety”** have highest importance
- These words indicate core domain relevance in meter testing remarks

✓ Observations:

- High TF-IDF score words are frequent yet contextually important.
- Domain-specific keywords are clearly identified.
- Model can use these features to differentiate between **departments/actions**.



Pending Work To Do

4. Model Development & Optimization:

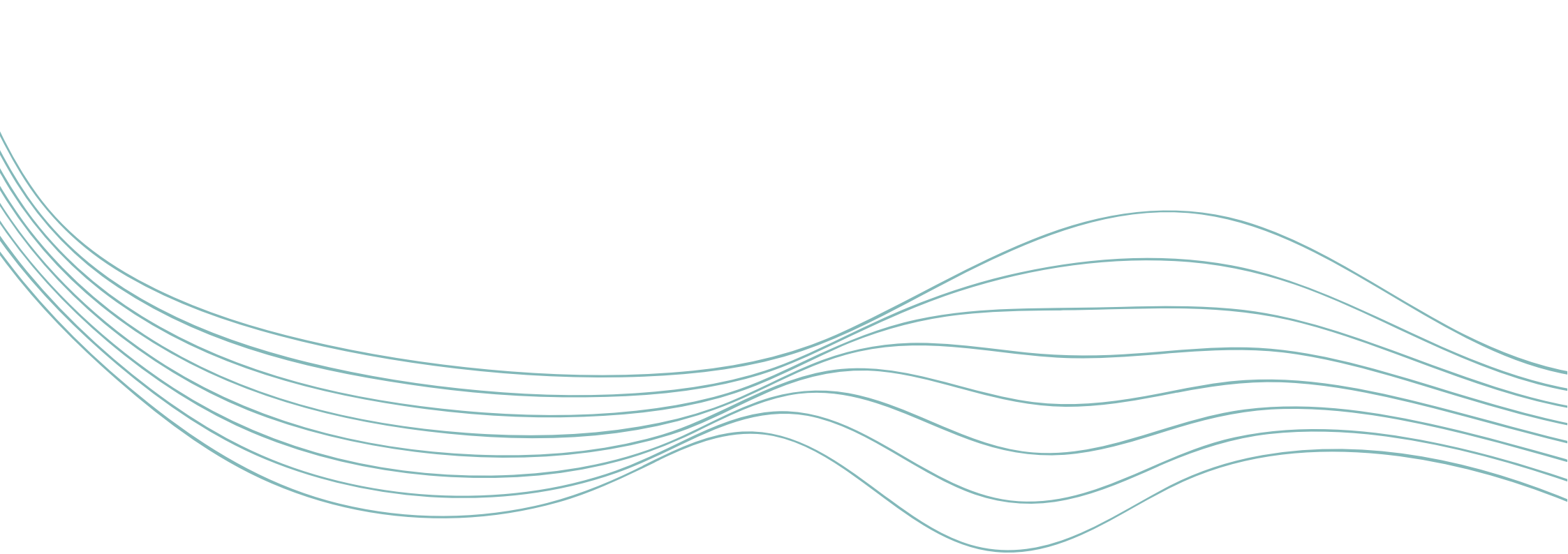
- Train **Logistic Regression model** on TF-IDF feature vectors
- Perform **train-test split** and evaluate model performance
- Use evaluation metrics:
 - ✓ Accuracy
 - ✓ Precision
 - ✓ Recall
 - ✓ F1-Score
 - ✓ Improve performance using:
- Improve performance using:
 - ✓ Hyperparameter tuning
 - ✓ Feature refinement (e.g., n-grams)

5. System Integration & Prediction Pipeline

- Develop an **end-to-end prediction system**
- Integrate:
 - ✓ Text preprocessing
 - ✓ TF-IDF vectorization
 - ✓ Trained classification model
- Enable system to:
 - ✓ Accept user remark as input
 - ✓ Predict concerned department in real-time

6. Live Deployment & User Testing

- Deploy model on **live/new dataset of meter testing remarks**
- Process real-time user inputs through the trained system
- Automatically classify remarks into respective departments
- Enable **direct routing of actions without manual intervention**
- Validate system performance with real-world data



THANKS

We are open for questions.

Cancellation_Hold Reason Code

Access / Availability Issues	
Code	Meaning
D030	Meter not traceable / wrong address
D120	Consumer not available
D315	Wrong pole number / wrong address
D385	No person at site / no phone response
D430	Premises Locked
D440	Premises Sealed
D500	Site under construction
D560	Visit not done
D610	Consumer refused testing

Replacement / Defective Meter Cases	
Code	Meaning
D255	Mechanical/Class-2 meter → Replace
D490	Display defect / RTC fail → Replace

Safety / Site Condition Issues	
Code	Meaning
D271	Unsafe site condition (damaged insulation, unsafe wiring)
D606	Water logging at site
D603	Space constraint / meter at excessive height

Electrical / Network Issues	
Code	Meaning
D395	Low voltage → Refer to Zone
D510	Solar / phase voltage issue
D330	Neutral not present from Supply End
D400	Solar Panel not working

Meter / Seal / Technical Blocking Issues	
Code	Meaning
D168	Meter terminal / box seals broken (suspected case)
D275	Meter box damaged
D320	Meter terminal seal missing → Refer to CEG
D460	Meter terminal & box seals missing → Refer to CEG
D601	Insufficient load for testing

Administrative / System Issues	
Code	Meaning
D270	Meter already replaced
D360	Meter retained as per system
D570	Work already done on another service order
D580	Wrong notification generated